Case Study - GTM Data Engineer

Objective

Our GTM team is struggling to keep track of customer sentiment across different platforms. We need a GTM Knowledge Base that indexes Gong Call Recording Transcripts and Slack Messages into a unified vector store to power an internal Deal Intelligence AI agent.

Your task is to design and partially implement the Ingestion & Transformation Engine for this knowledge base. You are free to make a decision on what type of architecture is the best choice for cost, speed, and complexity of system maintenance.

Guidelines

1. Raw Data Transformation

1.1 Data sources

You will have a few available data sources. Our data engineering team stores recordings conducted by [Gong]() between Sales, Customer Success, and Executive meetings with prospects and customers. The following schema is available in PostgreSQL-compatible database:

```python
GONG_CALL_SCHEMA = {
       "call_id": "string",
       "conversation_id": "string",
       "call_url": "string",
       "company_id": "string",
       "company_name": "string",
       "deal_id": "string",
       "deal_name": "string",
       "call_datetime": "datetime",
       "title": "string",
       "has_recording": "int",
       "has_transcript": "int",
       "transcript_json": "nullable(string)",
       "spotlight_brief": "nullable(string)",
       "spotlight_key_points": "nullable(string)",
       "spotlight_next_steps": "nullable(string)",
       "spotlight_outcome": "nullable(string)"
   }
```

HubSpot , our CRM, contains deal data is also available at:

1. Associations API — gets which meetings/notes belong to each deal:
    a. **POST** /crm/v3/associations/deals/{type}/batch/read (for "meetings" and "notes")
2. Batch Read API — fetches the actual content:
    a. **POST** /crm/v3/objects/meetings/batch/read with properties \
    b. **POST** /crm/v3/objects/notes/batch/read with properties

Lastly, Slack messages are available via:

1. Slack Messaging API
    a. **GET** https://slack.com/api/conversations.history

1.2 Goals:

- Extract & Standardize: Create a "Canonical Representation" of a customer interaction.
- Contextual Mapping: Programmatically link disparate logs (e.g., a Slack message from user_123 and a Gong call call_abc) to the same Hubspot Account ID.
- Output Generation: Produce a clean, structured JSON or Markdown payload that is optimized for LLM consumption.

2. System Design

- How would you store this data?
- Compare your chosen approach against the alternatives regarding latency, cost, and accuracy.
- GTM data changes hourly. How does your chosen architecture ensure the AI stays updated but also optimized for cost when refreshing data?
- How would you have users (sales, marketing, customer success team) interact with the AI agent so it is a fluid and fast experience for them

3. Reliability & Scale

- Describe how you would handle the ingestion of 1,000+ Gong calls per day. How do you ensure no data is dropped if an enrichment API rate-limits your requests?
- How would you reduce hallucinations?
- How would this agent interface with sellers and sales leaders?

Deliverables

Write a short report that has

1. A description of your approach
2. Flowchart diagram of your system architecture and explanation of why you made those choices
3. Provide us the script you built to transform the Gong transcript / Hubspot notes
4. Provide the cleaned version of the JSON dump (Gong/Hubspot)